Combining explicit Diophantine constructions for the consistency of ZFC

Comparison and improved constructions for Vladimir Reshetnikov

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Abstract

The eight repository packages do not have a single winner under every notion of simplicity. Combining their inputless compiler, ordinary strong Pell equation, strong-root elimination, and careful low-degree lifting gives several concrete improvements over the previous 15-variable, 67-operation, degree-155 polynomial and 34-variable quartic. We obtain a 14-positive-variable polynomial with 75 operations and degree 293; a 33-positive-variable quartic; a 25-variable sextic; and a 20-variable degree-eight equation. A 15-variable six-equation system retains the previous 61-operation cost while reducing maximum degree from 60 to 28. Positive-gap substitutions reduce that maximum further to 22 at a cost of two operations. A separate signed-domain argument yields smaller unrestricted-integer presentations.

The coefficient instance is also improved: direct blank-tape K215, restricted local windows, and tighter clause spacing reduce the radix from $2^{5^{67}}$ in package 06 to $2^{5^{43}}$, without changing the arithmetic skeleton. All four coefficients have bounded exact definitions; they are not expanded into decimal numerals. We distinguish mathematical proofs, imported Pell/compiler and machine theorems, finite independent checks, and the absent end-to-end Lean/Rocq certificate. No claim of global minimality or proof of Con(ZFC) is made.

1 Conclusions and the precise claim

Let $\mathbb{N} = \{0, 1, \ldots\}$ and let Con(ZFC) mean that no finite syntactic ZFC proof of a fixed contradiction exists in an adequate effective first-order calculus. The repository review is pinned to

`3b201cdea32d2c816f5f3d20734051f25fd3ef14`.

The eight packages were originally based on revision `cdd582bec13f908df3711fa9e05d1ddac368a5db`. Their provenance is preserved rather than treating the previous report as the source of every idea.

Theorem 1 (Mathematical construction)**.** *Fix the four integer coefficients β, K_0, M_C, M_F^* by Section 9. For each of the positive-domain polynomials or systems explicitly specified below, ZFC proves that its nonexistence of positive integer solutions is equivalent to* Con(ZFC)*. Replacing every positive variable v by $n_v + 1$ gives the same statement over $\mathbb{N}$, without changing the number of variables or the total degree.*

The proof composes the pinned K215 blank-halting calibration, the native-window/Pell compiler, and the reductions established here. It is an ordinary mathematical proof formalizable in ZFC, not a newly checked proof-assistant theorem or a supplied derivation encoded in ZFC syntax.

The following are distinct optimization choices, not claims that one row is universally simplest. Degree refers to a fully substituted polynomial, with fixed coefficients given weight zero.

Presentation	Variables	Equations	Operations	Degree
Previous compact polynomial	15	1	67	155
Previous comparison system	15	6	61	60
New ordinary comparison system	15	6	61	28
Its sum of squares	15	1	78	56
New ordinary product	15	1	69	99
New double-gap product	15	1	71	83
New double-gap system	15	6	63	22
Its sum of squares	15	1	80	44
New root-eliminated product	14	1	75	293
Its comparison system	14	5	69	244
New quadratic system	33	24	—	2
Its sum of squares	33	1	—	4
New cubic system	25	16	—	3
Its sum of squares	25	1	—	6
New degree-four system	20	11	—	4
Its sum of squares	20	1	—	8

All variables in this table are strictly positive. "Degree" for a system means its maximum residual degree. One binary addition, subtraction, or multiplication costs one operation, including a square and multiplication by a fixed numeral. Sharing and comparisons are free; coefficient bit length is not charged. The table therefore does not establish bit complexity or a shortest expanded formula.

Recommendation. For an inspectable short system, use the six ordinary equations: they improve degree without adding operations or unknowns. For the fewest explicit unknowns obtained here, use the 14-variable norm elimination. For a genuinely low-degree single equation, use the 33-variable quartic, or choose the 25-variable sextic or 20-variable degree-eight tradeoff. The 67-operation record among the compared compact polynomials is not improved. All these compatible forms can use the new, smaller coefficient tuple.

2 Comparison of the eight repository constructions

Table 1 compares representative presentations on their stated positive domain. Some packages contain several further alternatives; the accompanying comparison note records the distinctions and circuit filenames. Thirty literal arithmetic circuits were independently replayed as inert JSON. Downloaded repository programs were not executed.

Several apparent near-ties conceal real distinctions.

1. Package 02's normalized system has degree 60, while its ordinary-strong system has degree 28 at the same 63-operation cost. Package 03 has the analogous distinction at 64 operations. Package 05's degree-34 system instead unscales a normalized factor; that is not the ordinary replacement.

2. Packages 06 and 08 have identical 67-operation and 61-operation arithmetic skeletons up to one register rename. Their coefficient tuples are different and must not be mixed. Thus the independent inputless discoveries are credited to both packages, while package 06 supplies the smaller raw-machine coefficient baseline.

3. Root elimination in packages 03 and 05 reduces unknown count but greatly increases degree. This is the idea combined with the inputless skeleton below; the new contribution is the 14-variable specialization and its correctly ordered sign argument, not the general elimination trick.

4. Package 07's initially signed repunit, and package 02's squared marker, require existential reconstruction proofs. They are not free algebraic substitutions. The present report does not claim

Package	Representative metrics	Contribution used or contrasted here
01 explicit82	18/1/187/82; 20/9/42/72	Direct fixed-input specialization, retaining most variable-input infrastructure.
02 system63	16/7/28/63; 16/1/101/74	Ordinary strong equation; fixed marker; squared-budget variant.
03 system64	16/7/28/64; 15/1/295/82	Ordinary system and elimination of the strong Pell root.
04 poly73	16/1/156/73; 16/7/60/65	Affine marker factor, two-sided rather than folded machine.
05 fixed15 16	16/7/34/66; 15/1/295/82	Independent root elimination, outer-norm alternatives, signed-root proof order.
06 blank67	15/1/155/67; 15/6/46/62	Direct raw blank K215, smaller coefficients, modulo-five correction removed.
07 fixed70	16/1/169/70; 17/8/28/61	Signed repunit factor whose sign is recovered after power extraction.
08 start only67	15/1/155/67; 34/25/2/—	Independent inputless construction and compact quadratic lift.

Table 1: Metrics are variables / equations / maximum degree / operations. Different rows within a package need not be the same polynomial or the same coefficient recipe.

to have safely removed a further repunit variable using those ideas.

5. Variable counts over $\mathbb{Z}_{>0}$, $\mathbb{N}$, and $\mathbb{Z}$ are not interchangeable. In particular the previous 34-positive-variable quartic cannot be compared directly with a larger unrestricted-integer lift without charging the positivity conversion.

The machine alphabets also differ. Packages 01–05 use respectively 3510, 3502, 2628, 669, and 4376 cell symbols; package 06 uses 439; packages 07 and 08 use 4366. Package 04's small alphabet comes from a different two-sided compiler. Package 06's raw, padded-halting K215 machine avoids the normalization and input-erasing overhead. The improvement in Section 9 starts there rather than splicing its coefficients into an unrelated fixed-marker formula.

3 Shared inputless arithmetic

Use the fifteen positive unknowns

$$(J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma).$$

All symbols defined in the following display are polynomial abbreviations, not extra unknowns:

$$q = \beta J + 1, \qquad X = wq, \tag{1}$$
$$Y = sq^3, \qquad E = XY, \tag{2}$$
$$k = \eta + \zeta, \qquad c = kY + \eta, \tag{3}$$
$$a = E + Y, \qquad H = 4a + 3, \tag{4}$$
$$\Delta = a^2 + H, \qquad D = X + ac + \gamma H, \tag{5}$$
$$C = q - F - \alpha, \qquad L_0 = E(kY), \tag{6}$$
$$G = (q-1)(q-F) + \alpha = q(q-F) - C, \tag{7}$$
$$R = G(q^2 - 1) + (M_C + qM_F^*)J, \tag{8}$$
$$N_1 = \tau^2 - L_0(L_0 + k), \qquad N_2 = D^2 - \Delta c^2, \tag{9}$$

$$N_I = k - hE - R, \tag{10}$$
$$N_T = (K_0 + w)C + q - F - z(q-1), \tag{11}$$
$$V = c(Tf - 1) - Rf^2. \tag{12}$$

In these expressions K_0 is the fixed transport coefficient, not the temporary main-norm register denoted K in an older lift. Also M_F^* is the shifted source coefficient, not the native one-cell field mask.

For reference, the previous compact polynomial has

$$Q_n = (\Delta i c^2)^2, \quad N_A = Q_n(V^2 - y^2) + y^2, \quad N_S = \Delta f^2 - Q_n,$$
$$P_{67} = N_1 N_2 N_A N_I N_T N_S - \Delta. \tag{13}$$

Its six-equation presentation is

$$N_1 = N_2 = N_A = N_I = N_T = 1, \qquad N_S = \Delta.$$

A proof-only cancellation of $\Delta > 0$ precedes the integer-unit argument. One must not simply assert that all six factors of (13) are units: N_S equals Δ, not 1.

3.1 The ordinary strong replacement

Retain (1)–(12) and define

$$I = ic^2, \quad Q_o = \Delta(f^2 - 1), \quad N_A^o = Q_o(V^2 - y^2) + y^2, \quad N_S^o = 1 + I^2 - Q_o.$$

The recommended short system is

$$\boxed{N_1 = N_2 = N_A^o = N_I = N_T = 1, \qquad I^2 = Q_o.} \tag{14}$$

It uses 61 operations and has exact maximum degree 28. The polynomial

$$P_o = N_1 N_2 N_A^o N_I N_T N_S^o - 1$$

uses 69 operations and has exact degree 99. Its factor degrees are $22, 18, 28, 7, 2, 22$. The sum of the six squared residuals in (14) costs 78 operations and has exact degree 56.

This is not a same-i identity with the normalized system. Completeness sends i_n to $i_o = \Delta i_n$ and keeps the other coordinates. Soundness uses the ordinary strong-rank theorem; it does not assume $\Delta \mid i_o$.

3.2 Positive gaps lower degree

Set $\tau = L_0 + t$ and $y = V + e$, where t, e are new positive coordinates replacing τ, y, not additional variables. Then

$$N_1 = t^2 + L_0(2t - k), \qquad N_A^o = Q_o(-2Ve - e^2) + (V + e)^2.$$

The product has 71 operations and degree 83, while the six-equation system has 63 operations and maximum degree 22; its sum of squares has 80 operations and degree 44. The literal shared-square schedule, rather than a naively expanded display, defines the operation count.

On an old positive zero, $\tau^2 - L_0^2 = L_0 k + 1 > 0$, and the auxiliary norm gives $y^2 - V^2 = (V^2 - 1)/(Q_o - 1) > 0$. Thus $t, e > 0$. Conversely, on a positive sheared zero, the unit argument gives $Q_o = I^2 > 1$ and auxiliary norm 1. The restored $V + e$ is nonzero, so initially use $|V + e|$ to obtain an old positive zero. Its theorem gives $V > 1$, and hence $V + e > 0$. This establishes an actual positive-zero bijection, without assuming the needed auxiliary sign beforehand.

For exact degree, put $Q_0 = \beta J$, $k_0 = \eta + \zeta$ and $T_2 = w(Q_0 - F - \alpha) - zQ_0$. The sheared factor degrees are $12, 18, 22, 7, 2, 22$, and their product has nonzero leading form

$$16h\gamma i^2 w^6 s^{13} k_0^7 Q_0^{45} T f^3 e(2t - k_0) T_2.$$

This proves degree 83 uniformly for $\beta > 0$, rather than only by sampled evaluations.

4 Eliminating the strong root gives fourteen variables

Delete f from the supplied unknowns, retaining the other fourteen variables in Section 3. Keep all abbreviations not involving f and put

$$Q = (\Delta i c^2)^2, \qquad d = 1 + \Delta i^2 c^4, \tag{15}$$

$$b = c + Rd, \qquad u = cT, \tag{16}$$

$$A_* = Q(du^2 + b^2 - y^2) + y^2 - 1, \qquad B_* = 2Qub, \tag{17}$$

$$H_* = A_*^2 - dB_*^2. \tag{18}$$

The new single equation is

$$\boxed{P_{14} = N_1 N_2 N_I N_T (H_* + 1) - 1 = 0.} \tag{19}$$

It has 75 operations and degree 293. The equivalent positive-domain system is

$$\boxed{N_1 = N_2 = N_I = N_T = 1, \qquad H_* = 0.} \tag{20}$$

It has 69 operations and maximum degree 244. Its sum of squares has 82 operations and degree 488. The circuit evaluates dB_*^2 as $4(Qdu^2)(Qb^2)$, so the displayed auxiliary B_* is not an uncharged extra gate.

4.1 The order of the soundness proof

The delicate point is the sign of the recovered f. A rational square root is integral, but is not automatically positive. Nor may one invoke the original positive-auxiliary theorem while T or V is still signed. The following order avoids both gaps.

Unit signs and the boundary $C = 0$. At a positive zero of (19), all five factors are units. Since B_* is even, $H_* + 1 \equiv A_*^2 + 1 \pmod 4$, so it cannot be -1. Hence $H_* = 0$. The consecutive-product negative-Pell descent gives $N_1 = 1$; $\Delta \equiv 0$ or $3 \pmod 4$ excludes $N_2 = -1$. Thus $N_I = N_T = \epsilon \in \{1, -1\}$.

Transport gives $C \geq 0$, because

$$q - F - z(q-1) = 1 - F - (z-1)(q-1) \leq 0$$

and $K_0 + w \geq 2$ excludes $C < 0$. Do not yet assume $C > 0$. The native mask ranges and $M_F^* = M_F + \beta$ give

$$q \leq G \leq q^2 - q, \qquad 0 < (M_C + qM_F^*)J < (q-1)(1+2q), \tag{21}$$

$$q(q^2 - 1) < R < q^4 - q^3 + q^2 - 1 < q^4 - 2, \tag{22}$$

$$R > 3q + 1, \qquad q \geq 16, \qquad E \geq q^4 > R + 2, \quad a > R + 2. \tag{23}$$

These are pre-decoding bounds, valid even at $C = 0$. In particular $b, u, B_* > 0$ in the all-positive construction.

An integral root with no sign assumption. Set $f = A_*/B_*$. From $H_* = 0$, $f^2 = d \in \mathbb{Z}$. Reducing a rational fraction to coprime terms proves $f \in \mathbb{Z} \setminus \{0\}$. Put $V = uf - b$. The exact polynomial identity

$$QV^2 - (Q-1)y^2 - 1 = A_* - B_*f + Qu^2(f^2 - d)$$

yields the auxiliary norm, and $f^2 - \Delta i^2 c^4 = 1$ is automatic. For the next argument one may replace f by $|f|$ and T by $\operatorname{sign}(f)T$; Tf and all equations are unchanged. Now $f > 0$, but T may have either sign.

Signed auxiliary rank. Define Pell coordinates by

$$\chi_A(n) + \psi_A(n)\sqrt{A^2-1} = (A + \sqrt{A^2-1})^n.$$

Put $A_0 = a + 2$ and $P_0 = 2XY^2 + 1$. First and main norm classification gives

$$k = 2\psi_{P_0}(n), \quad \tau = \chi_{P_0}(n), \quad c = \psi_{A_0}(p), \quad D = \chi_{A_0}(p).$$

Here $p > n$, and the index congruence gives $2n \equiv R + \epsilon \pmod{E}$, so $n \geq (R-1)/2 \geq 24$. Pell growth gives $c > 2p$, while positive h gives $c > 2R$. The normalized strong norm gives

$$f = \chi_{A_0}(m), \qquad \psi_{A_0}(m) = ic^2.$$

Strong divisibility gives $p \mid m$. Expanding the unit to exponent m/p modulo c^2 then gives $c \mid (m/p)$; hence $c \mid m$ and $m > 2p$.

Let $\rho = \Delta i c^2$, so $Q = \rho^2$. Auxiliary norm classification gives

$$|\rho V| = \chi_\rho(\ell), \qquad y = \psi_\rho(\ell).$$

Divisibility of the first coordinate by ρ forces $\ell = 2v + 1$ odd. The integer polynomials Q_v defined by

$$\chi_Z(2v+1) = ZQ_v(Z^2)$$

satisfy

$$Q_v(-\Delta) = (-1)^v \psi_{A_0}(\ell), \qquad Q_v(0) = (-1)^v \ell.$$

Since $V \equiv -c \pmod{f}$, $V \equiv -R \pmod{c}$, and $\rho^2 \equiv -\Delta \pmod{f}$, squaring the first quotient congruence and using Pell doubling gives

$$\chi_{A_0}(2\ell) \equiv \chi_{A_0}(2p) \pmod{\chi_{A_0}(m)}.$$

The inherited plus step-down lemma with $0 < 2p < m$ gives $\ell \equiv \pm p \pmod{m}$, hence modulo c. The second congruence gives $R \equiv \pm p \pmod{c}$. Because $0 < R, p < c/2$, necessarily $p = R$. The possible sign of V is absorbed into these two plus/minus alternatives.

Writing $2n = R + \epsilon + vE$ with $v \geq 0$, the bounds $E > R + 2$ and $n < p = R$ exclude $v \geq 1$. If $\epsilon = -1$, then $p = 2n + 1$, and Pell duplication yields $c > k(Y+1)$, contradicting $kY < c < k(Y+1)$. Thus $\epsilon = 1$, $2n = R + 1$, and only now $N_T = 1$ excludes $C = 0$.

Power extraction and decoding before auxiliary positivity. Put $r = (R-1)/2$. The first/main Pell ratio gives

$$Y > X^r/3, \qquad a > X^{r+1}/3 > 2^R.$$

The main projection gives $X \equiv 2^R \pmod H$. Both positive representatives are smaller than a, so $X = 2^R$. Thus $q = 2^t$ for some integer t, and $R > 3q > 3t$ restores the symmetric q^3 scales. The ratio-only half-binomial calculation, which uses no auxiliary sign, yields

$$0 < \frac{c}{k} - \frac{\xi}{2} < \frac{16r}{X+1} < \frac{1}{2}, \quad \xi = \frac{(X+1)^{2r}}{X^r} = M + \theta,$$

$$M = \binom{2r}{r} + \sum_{j=1}^{r} \binom{2r}{r+j} X^j, \qquad 0 < \theta < \frac{1}{4}.$$

The integer M is even; the interval $Y < c/k < Y + 1$ forces $Y = M/2$. Central-binomial valuation gives $v_2(Y) = \mathrm{popcount}(r) - 1$, and therefore $\mathrm{popcount}(R) \geq 3t + 2$.

The exact repunit gives $q = B^N$, and inverse-population packing yields the two authentic shifted masks

$$(C-1)\ \&\ (M_C J + 1) = 0, \qquad F\ \&\ (M_F J - 1) = 0,$$

where M_F here is the native mask, not M_F^*. They type the Start-only word, put Start at its origin, and forbid selector 1. In particular they give $C \equiv 1 \pmod 4$, and only now $R \equiv 3 \pmod 4$. The physical transport quotient $z_{\mathrm{old}} = z - 1 + wC$ is positive. The unchanged compiler therefore decodes a genuine initialized halting K215 run. This proves soundness without asserting that the recovered root or auxiliary coordinate was positive.

Converse and optional positive restoration. An old positive P_{67} zero satisfies $f^2 = d$ and $A_* = B_* f$. Deleting f gives a P_{14} zero. Conversely, if a positive old-style tuple rather than just acceptance is desired, after decoding choose fresh

$$m = 2cR, \quad f' = \chi_{A_0}(m), \quad i' = \psi_{A_0}(m)/c^2,$$
$$\rho' = \Delta\psi_{A_0}(m), \quad V' = \chi_{\rho'}(R)/\rho', \quad y' = \psi_{\rho'}(R),$$
$$o' = (V' + c)/f', \qquad T' = (o' + Rf')/c.$$

Binomial expansion proves the c^2 divisibility. The established $R \equiv 3 \pmod 4$ gives the two minus congruences and integrality of o'; then $\gcd(f', c) = 1$ gives integrality of T'. All are positive. This is an existence map and may change i, T, y; it is not a same-tuple inverse.

4.2 Exact degree

With Q_0, k_0, T_2 as above, the four retained factors have exact degrees $22, 18, 7, 2$ and product leader

$$8h\gamma k_0^3 w^5 s^8 Q_0^{29} T_2.$$

The main factor's degree 18 includes the cancellation of its a^2c^2 terms. The degrees of Q, d, b, u are $46, 34, 38, 6$. The unique degree-122 term of A_* is Qb^2, so H_* has degree 244: the competing term dB_*^2 has degree 214. The degree-293 leader of P_{14} is

$$8h\gamma i^{12} k_0^{27} w^{21} s^{48} Q_0^{177} (Q_0 - F)^4 T_2.$$

For example the coefficient of $J^{182} h\gamma i^{12} \eta^{27} w^{21} s^{48} z$ is $-8\beta^{182} \neq 0$.

5 A quartic in thirty three positive variables

Use the fifteen original witnesses of Section 3, together with

$$N, E, P, c, a, X, D, C_2, G, L, R, F_2, V_0, V, g, I, B_a, \Delta.$$

These are eighteen additional unknowns, for a total of 33. In this section only $q = \beta J + 1$, $k = \eta + \zeta$, $H = 4a + 3$, $C = q - F - \alpha$ are abbreviations. Here is the entire system of 24 equations:

$$L = q^2 - 1, \qquad N = q(L + 1), \tag{24}$$
$$X = wq, \qquad a = E + sN, \tag{25}$$
$$E = X(a - E), \qquad c = k(a - E) + \eta, \tag{26}$$
$$P = E(c - \eta) + k, \qquad \tau^2 = P(P - k) + 1, \tag{27}$$
$$D = X + ac + \gamma H, \qquad \Delta = a^2 + H, \tag{28}$$
$$C_2 = c^2, \qquad D^2 - \Delta C_2 = 1, \tag{29}$$
$$k = R + 1 + hE, \qquad G = (q - 1)(q - F) + \alpha, \tag{30}$$
$$R = GL + (M_C + qM_F^*)J, \tag{31}$$
$$(K_0 + w)C + q - F - z(q - 1) = 1, \tag{32}$$
$$F_2 = f^2, \qquad V_0 = Tf - 1, \tag{33}$$
$$V = cV_0 - RF_2, \qquad g = y^2 - V^2, \tag{34}$$
$$I = iC_2, \qquad B_a = I^2, \tag{35}$$
$$B_a = \Delta(F_2 - 1), \qquad y^2 - B_a g = 1. \tag{36}$$

Each literal residual has degree at most two. Defining Q_4 as the sum of the squares of these 24 residuals gives one polynomial of exact degree four. This costs one fewer positive variable and one fewer quadratic equation than the previous 34-variable lift.

The changes are precise. Delete the old register $K = D^2 - 1 = \Delta C_2$, replace its two equations by $D^2 - \Delta C_2 = 1$, and replace $I = iK$ by $I = iC_2$. The latter changes normalized strong to ordinary strong, so this is not just algebraic register elimination. From a normalized solution use $i_o = \Delta i_n$; the values of I, B_a, f and every retained register stay unchanged.

5.1 Ordinary strong soundness and the missing divisibility

In the system, transport immediately gives $C > 0$. The pre-decoding estimates strengthen to

$$(2q - 1)(q^2 - 1) < R < q^4 - q^3, \qquad E > R + 2, \qquad a > R + 2.$$

First and main Pell classification gives $p > n \geq (R + 1)/2$, and elementary Pell growth then gives

$$c > (a + 2)\Delta^2, \qquad c > 2p, \qquad c > 2R.$$

The imported generic ordinary strong-rank theorem applies to

$$c = \psi_{a+2}(p), \qquad I = ic^2 > 0, \qquad I^2 = \Delta(f^2 - 1).$$

It yields

$$f = \chi_{a+2}(m), \quad I = \Delta\psi_{a+2}(m), \quad p \mid m, \quad c \mid m, \quad m \geq c > 2p.$$

In particular

$$f^2 - 1 = I\psi_{a+2}(m) = ic^2\psi_{a+2}(m),$$

so $c^2 \mid (f^2-1)$. This divisibility must not be obtained by informally dividing the original ordinary equation before the theorem is applied.

The supplied positive V now permits the integer restoration

$$o = cT - Rf, \qquad j = Tf - 1 - R(f^2-1)/c,$$

with $of = V + c > 0$ and $jc = V + R > 0$. Hence $o, j > 0$, and the auxiliary step-down applies. It gives $p = R$, and $E > R + 2$ excludes an index wrap. The remaining power, half-binomial, mask, and Start-only decoding arguments are unchanged. For the compact ordinary product, the unit-sign argument and the auxiliary gap estimate first recover these same positive coordinates; package 02 supplies precisely this ordinary-strong bootstrap.

The imported ordinary-rank statement is a mathematical theorem, not a new result of the finite JSON audit. This dependency is explicit in the artifact rather than silently replacing it by a claimed normalized identity.

6 A sextic in twenty five positive variables

Use the fifteen original witnesses with t in place of τ, and retain ten registers

$$E, c, a, D, L, R, V, I, B_a, \Delta.$$

Only $q = \beta J + 1$, $k = \eta + \zeta$, $H = 4a + 3$, $C = q - F - \alpha$ are abbreviations. The complete 16-equation system is

$$L = q^2 - 1, \tag{37}$$
$$a = E + sq(L+1), \tag{38}$$
$$E = wq(a - E), \tag{39}$$
$$c = k(a - E) + \eta, \tag{40}$$
$$t^2 + E(c-\eta)(2t-k) = 1, \tag{41}$$
$$D = wq + ac + \gamma H, \tag{42}$$
$$\Delta = a^2 + H, \tag{43}$$
$$D^2 - 1 = \Delta c^2, \tag{44}$$
$$k = R + 1 + hE, \tag{45}$$
$$R = L\big((q-1)(q-F) + \alpha\big) + (M_C + qM_F^*)J, \tag{46}$$
$$(K_0 + w)C + q - F - z(q-1) = 1, \tag{47}$$
$$V = c(Tf - 1) - Rf^2, \tag{48}$$
$$I = i(D^2 - 1), \tag{49}$$
$$B_a = I^2, \tag{50}$$
$$B_a = \Delta(f^2 - 1), \tag{51}$$
$$y^2 - B_a(y^2 - V^2) = 1. \tag{52}$$

All residuals have degree at most three. Their sum of squares Q_6 has exact degree six. This system retains the normalized strong equation, unlike the quartic in Section 5.

There is an exact positive-zero bijection with the previous 34-variable system. Eliminate

$$N = q(L+1),\ X = wq,\ C_2 = c^2,\ K = D^2 - 1,$$

$$G = (q-1)(q-F) + \alpha,\ F_2 = f^2,\ V_0 = Tf - 1,\ g = y^2 - V^2.$$

Also eliminate P and shear the first root by $\ell_0 = E(c-\eta)$, $t = \tau - \ell_0$, so $P = \ell_0 + k$. Because $a - E = sq^3 > 0$ and $c - \eta = k(a-E) > 0$, one has $\ell_0 > 0$; $\tau^2 = \ell_0(\ell_0 + k) + 1 > \ell_0^2$ gives $t > 0$. Conversely $\tau = t + \ell_0 > 0$ and $P = \ell_0 + k > 0$ restore the original equations. The old positivity lemma, applied to the restored supplied witnesses, gives the positivity of every deleted register. In particular $G > 0$ follows from transport, $V_0 > 0$ from the strong bound $f > 2c$, and $g > 0$ from the auxiliary gap.

The shear is necessary for the cubic count: eliminating P from the unshifted first norm would give a quartic. Eliminating both Δ and F_2 would also make the strong equation quartic, so Δ is retained. The first-norm residual's nonzero cubic leading form proves degree six for the sum of squares; leading forms of real squares cannot cancel identically.

7 A degree eight equation in twenty positive variables

Retain the original fifteen witnesses and only five registers E, c, a, V, I. Abbreviate the following polynomials, not extra variables:

$$q = \beta J + 1, \quad k = \eta + \zeta, \quad H = 4a + 3, \quad \Delta = a^2 + H, \quad C = q - F - \alpha, \quad R = k - 1 - hE.$$

The full system is

$$a = E + sq^3, \tag{53}$$
$$E = wq(a - E), \tag{54}$$
$$c = k(a-E) + \eta, \tag{55}$$
$$\tau^2 = E(c-\eta)\big(E(c-\eta) + k\big) + 1, \tag{56}$$
$$(wq + ac + \gamma H)^2 - \Delta c^2 = 1, \tag{57}$$
$$R = \big((q-1)(q-F) + \alpha\big)(q^2 - 1) + (M_C + qM_F^*)J, \tag{58}$$
$$(K_0 + w)C + q - F - z(q-1) = 1, \tag{59}$$
$$V = c(Tf - 1) - Rf^2, \tag{60}$$
$$I = ic^2, \tag{61}$$
$$I^2 = \Delta(f^2 - 1), \tag{62}$$
$$I^2(V^2 - y^2) + y^2 = 1. \tag{63}$$

All eleven residuals have degree at most four. This includes the degree-two abbreviation R in Rf^2; no hidden extra coordinate is being counted. Their sum of squares has exact degree eight, witnessed by $\beta^6 J^6 s^2$ from the first residual, since no other residual contains s.

This is exact register elimination from the 33-variable system, with a supplied elimination trace. Restoring all thirteen registers gives a positive solution: transport gives $C, G, R > 0$; the ordinary theorem gives $f > 2c$, hence $V_0 > 0$; and the auxiliary norm gives $g = y^2 - V^2 > 0$. In particular $V = 1$ is excluded by $V \equiv -c \pmod f$ with $2 < c < f/2$. Thus the projection and restoration are inverse positive-zero maps. Its dependence on the ordinary strong-rank theorem is inherited from the 33-variable source.

8 Domains and unrestricted integer equations

A sum of squared integer residuals vanishes exactly when every residual vanishes. This justifies all the single-polynomial conversions above over $\mathbb{Z}_{>0}$ or $\mathbb{N}$; the shift $v = n_v + 1$ preserves degree. It does not justify forgetting positivity and allowing arbitrary signed variables in the compact products.

A separate signed-domain argument shows that only these nine original ports must remain positive:

$$J, F, \alpha, z, i, s, w, \eta, \zeta.$$

The six remaining supplied coordinates f, h, T, τ, y, γ may be signed in the normalized 15-coordinate construction. All derived registers may be signed as well when their defining equations are retained. The statement is equality of existential truth values, not a pointwise absolute-value bijection.

The proof first derives $C \geq 0$ and (22). From the index equation, $h \geq 0$; $h = 0$ would put the first Pell coefficient k strictly between 2 and XY^2, where a consecutive-square argument excludes it. Thus $h > 0$. First, main, strong, and auxiliary Pell classification then works with absolute roots. The signed quotient argument in Section 4 gives $p = R$ without any assumption on the sign of T or V.

The sign of the main root is not skipped. Write $A = a + 2$ and $e_R = \chi_A(R) - a\psi_A(R)$. Projection gives $e_R \equiv 2^R \pmod{H}$. If $D = -\chi_A(R)$, the defining equation for D, multiplied by the identity

$$(\chi_A(R) - ac)(\chi_A(R) + ac) = 1 + Hc^2,$$

forces $X2^R + 1 \equiv 0 \pmod{H}$. The ratio bound is strong enough to give $0 < X2^R + 1 < a + 1 < H$, a contradiction. Hence $D > 0$, then $X = 2^R$, and $e_R = 2c - \psi_A(R - 1) > c > X$ gives $\gamma > 0$. Decoding now proceeds with no sign assumptions on the auxiliaries. The ordinary construction uses its generic strong-rank theorem to supply the same divisibility and bounds.

Define

$$\mathrm{Pos}(u, v, r) = 1 + u^2 + v^2 + r(r + 1).$$

It is always positive and is surjective onto $\mathbb{Z}_{>0}$. Indeed $4n - 3 \equiv 1 \pmod 4$ is never of the excluded form $4^a(8b + 7)$, so Legendre's three-square theorem expresses it as one odd square and two even squares; division by four gives the displayed formula. Replace each of the nine positive ports by its own copy of Pos. This adds eighteen variables, not three variables for every coordinate.

For the root-eliminated construction, only five of the remaining coordinates are signed. Its signed-domain extension also requires ruling out $T = 0$, so that division by B_* remains legal. At $T = 0$, $u = 0$, $H_* = 0$ forces $A_* = 0$ and thus the auxiliary norm with $|V| = b$. The pre-decoding bounds give $1 < b < Q$, while the nontrivial auxiliary Pell gap gives $b \geq 2Q - 1$, a contradiction. This closes the denominator case rather than assuming it away.

Unrestricted integer presentation	Variables	Equations	Operations	Degree
Root-eliminated product	32	1	138	584
Normalized compact product	33	1	130	304
Ordinary double-gap product	33	1	134	159
Ordinary quadratic lift with positivity relations	60	33	—	2
Its sum of squares	60	1	—	4

The ternary positivity prefix uses seven operations per positive port, hence adds 63 operations to each compact circuit. The 60-variable system instead retains the 33 original coordinates and adds 27 variables and nine quadratic relations $v = \mathrm{Pos}(u_v, v_v, r_v)$.

The degree counts use weights two for the nine positive ports and one for all remaining coordinates, before substitution. The factor weighted degrees of the 14-variable product are $44, 35, 13, 4, 488$, summing to 584. For the ordinary double-gap product they are $24, 35, 39, 13, 4, 44$, summing to 159. The latter is not twice degree 83: its gap roots remain signed and have weight one. The weighted leading forms are nonzero at the actual $\beta > 0$. Disjoint positivity blocks preserve this fact, since setting each block's last two coordinates to zero leaves the injective substitution $v \mapsto u_v^2$ on leading forms. Thus these are exact degrees, not only substitution upper bounds.

Further shipped tradeoffs include 47 unrestricted variables at degree eight, 43 at degree ten, and 38 at degree sixteen. Their literal residuals and independent receipts are supplied, but the table highlights the clearest choices rather than treating every elimination as a separate main result.

9 A smaller fully specified coefficient instance

This section defines fixed integers, not existential parameters. No step waits for K215 to halt. The literal 430-transition machine table in the archive has SHA-256

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

The first action is $(0,0) \mapsto (1,\mathrm{R},93)$; the final action $(68,1) \mapsto (1,\mathrm{R},215)$ retains both its write and move. Only the newly added transitions $(215,s) \mapsto (s,\mathrm{stay},215)$ are stationary. The raw machine is calibrated to finite ZFC refutation by the pinned upstream theorem.

9.1 The 439 cell codes and the local language

Use codes 0 and 1 for headless tape symbols, $2(q+1)+s$ for a head in state $0 \le q \le 215$ on symbol $s \in \{0,1\}$, and codes 434,435,436 for phases L, Q, R with respective payloads blank/headless, blank/state 0, blank/headless. Codes 437 and 438 are vertical and horizontal boundaries, denoted V_b, H_b; they have no payload or phase.

For a 3×3 row-major window, write P, L_c, C_c, R_c, N for its prior, left, center, right, and next cross entries. Time increases downward. A window is *Good* precisely when these finite tests hold:

1. Vertical-boundary membership agrees at C_c, N; C_c, R_c are not both V_b; C_c, N are not both H_b. If neither C_c nor R_c is V_b, their horizontal-boundary memberships agree.

2. A boundary center has no further original center conditions. At an interior center, a phase is present exactly when $P = H_b$. When a phase is present, a V_b on the left forces L, and one on the right forces R; unless $R_c = V_b$, the adjacent phase pair is one of LL, LQ, QR, RR.

3. An interior head with $N = H_b$ is in state 215. A head may not move through an adjacent V_b.

4. At an interior center, if $N \neq H_b$, its payload is the exact deterministic update of the current three payloads. A center head writes; otherwise the symbol stays. Incoming heads come from a right-moving left head, stationary center head, or left-moving right head. More than one incoming head is forbidden. A vertical boundary is blank and headless for this test; a horizontal neighbor cannot supply an interior update.

5. In each of the window's three rows, at most one entry is headed. There are 433 headed codes and six headless codes. This is the added restriction relative to package 06.

These are asymmetric center tests, not extra unmentioned tests centered at all nine cells. The first four clauses are package 06's compiler; the last clause is the new language restriction.

Assign selectors 0 and 1 to the following windows:

$$S = (438,438,438;\ 434,435,436;\ 0,1,188),$$

$$E_b = (0,438,0;\ 0,437,438;\ 0,437,0).$$

Semicolons separate rows. Both are locally Good. The true first action of K215 determines the lower row of S in a valid plane. The window E_b cannot occur in a globally overlapping plane: its prior center is H_b and center is V_b, contradicting the vertical-boundary propagation test centered one row earlier.

Put $W_0 = S$, $W_1 = E_b$, and enumerate all other Good windows lexicographically. This bounded enumeration of 439^9 tuples defines exactly

$$\kappa = 10{,}762{,}728{,}984{,}864$$

windows. Selector 1 can therefore be forbidden without removing any globally valid computation.

9.2 Why the extra head restriction preserves halting

Soundness follows from the original local conditions. More explicitly, overlapping windows with cyclic offsets -1 and $-h$ reconstruct a plane by taking the center at cyclic index $i-d_x-hd_y$ for offset (d_x, d_y). This remains valid if offsets coincide. Horizontal periodicity and the phase graph LL, LQ, QR, RR force vertical boundaries on both sides of any occurrence of S, because that graph has no cycle through Q. Vertical propagation gives a finite strip. The horizontal row above Q propagates across it, and temporal periodicity gives a next such row. Between them lies one blank initialized computation with a unique state-0 head; the update and no-escape rules preserve it, and the last interior head must be in state 215.

Completeness needs a seam argument, not just "a machine has one head." Given a finite halting run, choose a strip with its leftmost and rightmost two interior cells outside every visited head position. Pad with absorbing halt rows, and let the finite rectangle be G of width w_b and height H_t. Extend it helically by

$$\mathcal{T}(x,t) = G\big(x \bmod w_b,\ (t + \lfloor x/w_b \rfloor) \bmod H_t\big).$$

Within a strip every three-cell row has at most one head. A three-cell row crossing a helical seam meets the boundary or only its permanently headless two-cell margins. It cannot collect heads at two different time offsets. Thus the added restriction holds everywhere. The periods are $(w_b, -1)$ and $(0, H_t)$; cyclic coordinate

$$i = -(x - x_S) - w_b(t-1) \pmod{w_b H_t}$$

places S at zero and gives right/next-time offsets $-1, -w_b$. Width and height can independently be padded to arbitrarily large powers of five, as required by the compiler's Boolean dummy controls.

The exact window count is independently reproduced by two methods. For a headed row center the corners have $6^2 = 36$ assignments; for a headless center they have $6^2 + 2 \cdot 6 \cdot 433 = 5232$. The original predicate depends only on the cross, so these weights give an exact finite count. The C++ exhaustive center/pair calculation and Python category calculation agree for each of the 439 centers, not merely in their grand total.

9.3 The complete bounded coefficient recipe

Set $a_b = 439$, $\ell = 44$, $A_b = 2^\ell$ and

$$p_b = \max\left(0, \left\lceil \frac{\kappa - 2\ell - 21a_b - 4}{2} \right\rceil\right), \qquad j_{\max} = 9a_b + 4 + p_b,$$

$$\mu = A_b - 2 + \sum_{j=1}^{j_{\max}} A_b^j, \qquad m_0 = 2\ell + 30a_b + 8 + 2p_b,$$

$$n_{\mathrm{pos}} = m_0 + 1, \qquad n_{\mathrm{dum}} = n_{\mathrm{pos}} - \kappa - 9a_b - 4,$$

$$e_* = \kappa + 12a_b, \qquad M_0 = \kappa + 15a_b + n_{\mathrm{dum}}.$$

For $0 \le r, c \le 2$ and $0 \le u < a_b$, define

$$e(r,c,u) = \kappa + 3a_b + (2-r)3a_b + (2-c)a_b + u, \quad j(r,c,u) = 1 + (3r+c)a_b + u.$$

The disjoint native exponent set is

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, \kappa - 1\} \ \cup\ \{e(r,c,u)\} \\ & \cup\ \{e_*, \dots, e_* + n_{\mathrm{dum}} - 1\} \ \cup\ \{M_0, 3M_0, 9M_0, 27M_0\}.\end{aligned}$$

It has n_{pos} entries. Assign the following coefficient weights:

$$c_i = 1 + \sum_{r,c=0}^{2} A_b^{j(r,c,W_i[3r+c])} + \sum_{t=1}^{4} A_b^{9a_b+t} \qquad (0 \le i < \kappa),$$
$$c_{e(r,c,u)} = A_b^{j(r,c,u)},$$
$$c_{3^{t-1}M_0} = A_b^{9a_b+t} \qquad (1 \le t \le 4),$$

and $c_e = 0$ at the dummy exponents. The next layout values are

$$E_{\max} = 27M_0, \qquad H_C = E_{\max} + 24M_0 + 3a_b + 1,$$
$$T_1 = H_C + 2E_{\max} + a_b + 1, \qquad \delta_T = 4\lceil E_{\max}/2 \rceil + 2,$$
$$T_2 = T_1 + \delta_T.$$

Let L_b be the least power of five at least $T_2 + 2E_{\max} + 2$, and put

$$U = 4(n_{\mathrm{pos}} + 2)\big(28(\kappa + 9a_b + 4) + 6\big) + 8,$$
$$b_{\min} = \max\{\ell(j_{\max} + 1) + 3,\ \ell(9a_b + 4) + \mathrm{bits}(U) + 1,\ 4\},$$

where $\mathrm{bits}(U) = \lfloor \log_2 U \rfloor + 1$. Let b_b be the least power of five at least $b_{\min}$, and define

$$\rho_b = 2^{b_b}, \qquad d_b = b_b L_b, \qquad B = \rho_b^{L_b}.$$

Finally set

$$D_C = \rho_b^{3a_b} + \rho_b^{H_C + a_b} + \rho_b^{8M_0} + \rho_b^{24M_0} + \sum_{e \in \mathcal{E}} c_e(\rho_b^{T_1 - e} + \rho_b^{T_2 - e}),$$
$$D_R = \rho_b^{H_C},$$
$$M_{F,0} = \mu(\rho_b^{T_1} + \rho_b^{T_2}) + \sum_{\substack{0 \le r \le 1 \\ 0 \le c \le 2 \\ 0 \le u < a_b}} \rho_b^{e(r,c,u)} + \sum_{\substack{0 \le r \le 2 \\ 0 \le c \le 1 \\ 0 \le u < a_b}} \rho_b^{H_C + e(r,c,u)} + \rho_b^{9M_0} + \rho_b^{27M_0},$$
$$M_C = B - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \ne 1}} \rho_b^{e} - 2\rho_b^{e_*}, \qquad M_F = M_{F,0} + 4.$$

The four actual coefficient ports used everywhere in this report are exactly

$$\boxed{\beta = B - 1, \qquad K_0 = D_C + BD_R, \qquad M_C, \qquad M_F^* = M_F + B - 1.} \tag{64}$$

No coefficient is a free machine index, a proof oracle, or a halting-dependent value. The exponents and finite sums specify genuine integers. They are not a claim that those integers have been feasibly expanded.

9.4 Concrete layout and why it remains valid

The new scalar values include

$$p_b = 5{,}381{,}364{,}487{,}777, \qquad j_{\max} = 5{,}381{,}364{,}491{,}732,$$
$$n_{\mathrm{pos}} = 10{,}762{,}728{,}988{,}821, \qquad n_{\mathrm{dum}} = 2,$$
$$M_0 = 10{,}762{,}728{,}991{,}451, \qquad E_{\max} = 290{,}593{,}682{,}769{,}177,$$
$$H_C = 548{,}899{,}178{,}565{,}319, \qquad T_1 = 1{,}130{,}086{,}544{,}104{,}113,$$
$$T_2 = 1{,}711{,}273{,}909{,}642{,}471.$$

In particular

$$b_b = 5^{21}, \qquad L_b = 5^{22}, \qquad d_b = 5^{43}, \qquad B = 2^{5^{43}}.$$

Package 06 used $b_b = 5^{33}, L_b = 5^{34}, B = 2^{5^{67}}$. The bit length of $B-1$ has decreased by the factor 5^{24}. It remains astronomically large.

The clause bands need only

$$\delta_T > 2E_{\max}, \quad L_b - \delta_T > E_{\max}, \quad T_2 + E_{\max} < L_b.$$

The selected gap is the least integer above $2E_{\max}$ congruent to 2 modulo 4, and the chosen L_b ensures the remaining inequalities. The bounded mass estimates

$$\sum c_e \le 14(\kappa + 9a_b + 4)A_b^{9a_b+4}, \qquad \mu < A_b^{j_{\max}+1}$$

ensure the inherited no-carry margin by the chosen b_b. All non-anchor exponents lie below $M_0 - 3a_b$. Shifts $3a_b$ and $H_C + a_b$ perform the copy tests; $8M_0, 24M_0$ perform exactly the two anchor tests. There is no other support pair of difference $18M_0$. The separated clause bands introduce no cross-contributions.

The mask populations sum to d_b, $M_C \equiv 2 \pmod 4$, and $M_F \equiv 4 \pmod 8$. Because b_b is a power of five, $\rho_b \equiv 2 \pmod 5$; paired clause terms cancel as $\delta_T \equiv 2 \pmod 4$. The remaining terms give

$$2D_C - D_R \equiv 3 \pmod 5.$$

No conditional correction bit or fifth coefficient is needed. These support and congruence arguments preserve the native compiler interface, including its upper dummy bit. They justify using the new tuple in all compatible arithmetic forms rather than merely replacing a large numeral by a smaller guess.

10 What was verified and what remains imported

The archive separates three kinds of evidence.

1. **Mathematical reductions.** The signed-root argument, ordinary-strong transfer, positive-gap bijections, register eliminations, domain refinements, local head restriction, helical completeness, and shorter-band support argument are supplied in readable proof notes as well as this report.

2. **Finite reproducible checks.** New local programs parse received files only as inert data. They verify circuit dependencies, liveness, operation ledgers, signed polynomial identities, sparse elimination identities, exact low-degree residuals, machine-table identity, exact local-window counts, and small finite support layouts. Finite-field line tests diagnose the stated high degrees; the uniform nonzero leading forms in the proofs, not those tests alone, establish exact degree.

3. **Imported mathematical theorems.** K215 blank halting iff a finite ZFC refutation; Pell classification, strong divisibility, ordinary strong rank and signed plus step-down; first/main ratio and half-binomial valuation; inverse-population packing and the native compiler. Their source locations and pins are retained. No finite sample is claimed to prove an arbitrary halting, Pell-rank, or compiler theorem.

The coefficient audit agrees on all 430 machine transitions, all 439 per-center window counts, six sparse support layouts with 336,576 basis-convolution checks, and 75 padded halting-control helices with 13,250 local checks. The control helices are finite test machines with the actual K215 first action, not an accepting K215 run. No ZFC refutation, giant expanded coefficient tuple, or new proof-assistant certificate is produced.

The low-degree audits independently expand every residual, replay every elimination, check positive-lift reasoning, and verify exact degrees after the $v = n_v + 1$ shift. The degree-four and degree-six claims are not degree bounds of an unexpanded circuit with uncounted registers: their JSON contains literal polynomials in exactly the listed variables.

No consistency assumption is required for the equivalence in Theorem 1. The theorem does not prove either side. If ZFC is consistent, its own consistency and hence the corresponding nonexistence statement cannot be proved in ZFC by the usual second incompleteness theorem. It would be incorrect to claim full two-sided independence from consistency alone without the additional hypotheses needed for the other direction.

10.1 Reproducible package

The attached ZIP contains the editable LaTeX source, compiled PDF, exact JSON circuits and residual lists, the literal machine table, bounded coefficient specification, readable proof notes, source pins, check receipts, and newly authored check programs. A package manifest records SHA-256 hashes. Reproduction instructions state which files are source data and which programs were authored for this audit; no downloaded repository script needs to be executed.

Independent mathematical review found no remaining gap in the new ordinary-rank transfer, signed-root proof order, signed-port reduction including the main-root sign, denominator exclusion, low-degree positive reconstruction, or the double-gap bijections. A separate replay agreed with the compiler counts and support checks, the compact circuit ledgers, and the low-degree and integer-domain audits. These are human-readable proof review and finite computation, not a kernel-checked composite proof.

From the unzipped package, run `python3run_checks.py` for the standard-library replay, or add `--exhaustive` to compile and run the separately authored C++17 local-window counter. The runner verifies hashes and source pins, works in a temporary copy, and executes only the new audit programs. It does not import or execute the received repository programs. The final PDF was rendered and every page inspected; the rendering receipt and manifest identify the reviewed artifact.

11 Further directions that are worth testing

The following are research questions, not advertised improvements.

1. Can the signed repunit idea of package 07 eliminate another inputless coordinate while retaining the asymmetric $X = wq$ bootstrap? Its $X = wq^2$ proof cannot simply be transplanted. A successful proof would need pre-decoding packing bounds and a delayed sign recovery.

2. Can strong-root elimination be combined with an ordinary strong certificate to beat the present variable/degree tradeoff? Squaring a rational reconstruction without controlling its denominator and signed auxiliary rank is not enough.

3. Can the nine genuinely positive ports be reduced further? Each reduction saves two unrestricted-integer variables under the three-variable positivity parametrization. The signs of the data/field budget and scale ports are structural, so every proposed relaxation needs a soundness proof.

4. Can one remove another register from the 33-variable quadratic system without creating a cubic residual or sacrificing a necessary positivity argument? The 25-variable and 20-variable systems show the cost of moving along this tradeoff.

5. Can the local language be pruned by additional globally necessary restrictions while preserving all helical seams and independently paddable widths and heights? A concrete next test is consistency

of the relative positions and states of observed heads in consecutive window rows with the actual transition, excluding initialization boundaries. The two-cell seam margins may support such a restriction, but it has not been implemented or claimed here. Exact counting alone does not establish the completeness of a pruning rule.

6. The most valuable confidence improvement is an end-to-end formalization of the specific compiler interface and new reductions. Existing K215 endpoint certification does not automatically certify this Diophantine bridge.

7. Classical universal-polynomial results give fewer unknowns in existential representations, including nine-variable bounds. Extracting a competitive, completely instantiated coefficient recipe and arithmetic circuit is a different task. The present fourteen-variable result is not a global record.

Sources and attribution

The primary immutable collection is VladimirReshetnikov/ProveIt, Con(ZFC) Diophantine forms at the reviewed revision. The archive's source-pins file gives a verified URL and content hash for every downloaded source used.

The principal mathematical comparisons use package 02's `FIXED_INPUT75_ORDINARY_PROOF.md`; package 03's `proofs/FIXED_MARKER.md` and `proofs/DEGREE_295.md`; package 04's report; package 05's `proofs/fixed_input_proof_notes.md`; package 06's raw blank-helical and Pell-audit notes; package 07's fixed-input notes; and package 08's inputless polynomial and 34-variable lift. The new formulas combine these ideas and are not attributed to a single preceding report.

The positivity parametrization uses Legendre's three-square theorem; a machine-checked version of the representation result is documented in the Archive of Formal Proofs entry Three Squares. This citation does not certify the new Diophantine construction.